The Interaction Between Synaptic Noise and Intrinsic Excitability in Representational Drift

Computational Neuroscience — Final Project

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Abstract

Hebbian plasticity encodes memories as weight changes in recurrent networks, yet memories remain vulnerable when network activity is suppressed by anesthesia or deep inhibition. We implement a minimal rate-based recurrent network ($N = 10$) with an all-inhibitory structural baseline ($W_{\text{struct}} \leq 0$), Hebbian weight plasticity, per-neuron intrinsic excitability (ε_i), and homeostatic mean-rate regulation, and simulate a three-day equal-duration protocol (20 s each) across four conditions: a *Baseline* (no noise), *Condition A* (synaptic noise), *Condition B* (slow intrinsic excitability fluctuations), and *Condition C* (both). All conditions share an identical encoding phase (Day 1, 20 s). During suppression (Day 2, 20 s, $I_p = -3$), the ReLU nonlinearity silences Hebbian updates; the trace decays passively to $\approx 37\%$ of its Day 1 value. On Day 3 (20 s), homeostasis overcomes the suppressive input and restores firing. The key finding is a *stochastic bifurcation* introduced by excitability fluctuations: Baseline and Condition A produce a deterministic, reproducible recovery ($m_{\text{mem}} \approx 1.1$, std ≈ 0), whereas Conditions B and C show high outcome variability (std ≈ 0.74) across random seeds—the network can converge to either the correct memory attractor or a spurious one depending on the specific excitability trajectory. Synaptic noise alone does not alter mean recovery or its variance.

1 Introduction

Cortical circuits must encode memories during learning while keeping those memories stable during retrieval. Hebbian plasticity—the strengthening of connections between co-active neurons—provides a mechanism for encoding [Rule et al., 2019], but the stored trace is vulnerable to any perturbation that silences activity: anesthesia, deep sleep, or strong inhibitory drive halts potentiation and allows passive weight decay toward the structural baseline.

Whether a memory trace survives suppression, and whether it can subsequently be "rescued" by homeostatic regulation, depends on at least two factors that have not been studied jointly: (1) synaptic noise, which adds stochasticity to weight dynamics and may erode the trace even when activity is suppressed; and (2) slow fluctuations in intrinsic excitability [Delamare et al., 2023], which shift individual neurons' thresholds for reactivation and may bias the pattern of recovery toward or away from the encoded assembly.

Here we implement a minimal rate-based network and ask: how do synaptic noise and excitability fluctuations, separately and jointly, affect the integrity of a Hebbian memory trace and the fidelity of its homeostatic rescue, and how reproducible are these effects across random noise realizations?

2 Methods

2.1 Network model

We simulate $N = 10$ rate units governed by

$$\tau_r \, \dot{r}(t) = -r(t) + W(t)\, \phi(r(t)) + I(t) + \varepsilon + h, \tag{1}$$

where $\phi = \mathrm{ReLU}$, $\varepsilon \in \mathbb{R}^N$ is a per-neuron excitability bias (zero in Baseline and A), and h is a scalar homeostatic drive (active only on Day 3). The structural weight baseline W_{struct} is entirely inhibitory: all entries drawn from $-\mathrm{Uniform}(0, W_s/N)$, diagonal set to zero. Recurrent weights evolve by

$$\tau_W \, \dot{W}(t) = -\lambda\big(W(t) - W_{\text{struct}}\big) + \eta \, \phi(r) \, \phi(r)^\top, \tag{2}$$

and the homeostatic variable follows

$$\tau_h \, \dot{h}(t) = \kappa\big(m^* - m_{\text{mean}}(t)\big), \qquad m_{\text{mean}} = \tfrac{1}{N}\mathbf{1}^\top r. \tag{3}$$

All parameters are in Table 1. Equations are integrated with simultaneous Euler steps of $dt = 10\,\text{ms}$: $\dot{r}$ and $\dot{W}$ are computed from the *same* $r(t)$, $W(t)$ before any update.

Symbol	Meaning	Value
N	Number of neurons	10
τ_r	Rate time constant	10 ms
τ_W	Weight time constant	2000 ms
τ_h	Homeostasis time constant	500 ms
η	Hebbian learning rate	0.005
λ	Weight decay rate	0.1
κ	Homeostatic gain	0.2
m^*	Target mean rate	0.5
W_s	Structural weight scale	0.1
dt	Euler step size	10 ms

Table 1: Model parameters.

2.2 Three-day protocol

All three days have equal duration of 20 s (2000 Euler steps each).

Day 1 — encoding (20 s). A sparse cue $I_{\text{base}} \in \mathbb{R}^N$ activates five neurons (indices $\{0, 3, 4, 5, 7\}$, amplitudes 0.85–1.48) for 20 s. No homeostasis; $\varepsilon = 0$. Hebbian plasticity accumulates a memory trace $\Delta W = W - W_{\text{struct}}$. The encoding is identical across all four conditions.

Day 2 — suppression (20 s). A uniform negative input $I_p = -3\,\mathbf{1}$ drives all rates below zero, silencing the Hebbian term via ReLU. Weights decay passively toward W_{struct} by the factor $\exp(-\lambda t/\tau_W) = e^{-1} \approx 0.37$. Homeostasis is off.

Day 3 — homeostatic rescue (20 s). I_p *remains active* while homeostasis turns on. The drive $h(t)$ rises to overcome the suppressive input and restore mean firing rates. Conditions differ by the presence/absence of synaptic noise and excitability fluctuations during Days 2 and 3.

2.3 Four conditions

All conditions start Day 2 from the identical Day 1 final state (W_{d1}, r_{d1}).

- **Baseline**: no noise ($\sigma_\eta = 0$), $\varepsilon = 0$.
- **A – synaptic noise**: each weight update includes an i.i.d. Gaussian term, so Eq. (2) becomes $\tau_W \, \dot{W} = -\lambda(W - W_{\text{struct}}) + \eta \, \phi(r)\phi(r)^\top + \sigma_\eta \, \xi(t)$, with $\sigma_\eta = 0.01$ and $\xi_{ij}(t) \sim \mathcal{N}(0, 1)$ i.i.d.; $\varepsilon = 0$.
- **B – excitability**: each ε_i follows an Ornstein–Uhlenbeck process

$$\tau_\varepsilon \, d\varepsilon_i = -\varepsilon_i \, dt + \sigma_\varepsilon \sqrt{2\, dt/\tau_\varepsilon} \; \xi_i(t), \tag{4}$$

  with $\tau_\varepsilon = 1000\,\text{ms}$, $\sigma_\varepsilon = 0.8$, $\xi_i \sim \mathcal{N}(0, 1)$; no synaptic noise.

- **C − both**: synaptic noise *and* excitability fluctuations as above.

2.4 Analysis

The primary readout is the memory projection $m_{\text{mem}}(t) = r_m \cdot r(t)/\|r_m\|$ where r_m is the Day 1 final pattern. We also track the homeostatic drive $h(t)$, the mean firing rate $m_{\text{mean}}(t)$, and the Hebbian trace norm $\|\Delta W\|_1/N^2$.

To assess reproducibility, we run each condition with $n = 20$ independent random seeds (controlling the noise and OU trajectories in Days 2–3 only; Day 1 encoding is always deterministic). We report mean ± std of m_{mem} at the end of Day 3.

3 Results

3.1 Day 1: Hebbian weight buildup from an all-inhibitory baseline

During the 20 s encoding period, the five active neurons sustain firing rates near $\bar{r} \approx 0.77$; the remaining five are silent (Figure 1a). Because $\tau_r \ll \tau_W$, rates reach quasi-steady state almost instantly and Hebbian plasticity slowly accumulates the outer product $\phi(r)\phi(r)^\top$.

Figure 1b–d shows $\Delta W = W - W_{\text{struct}}$ at $t \approx 7$, 13, and 20 s. The weight change is approximately rank-1: a block of excitatory entries grows among the five co-active neurons while all other pairs remain in the inhibitory structural baseline (see also Figure 4, Day 1 column).

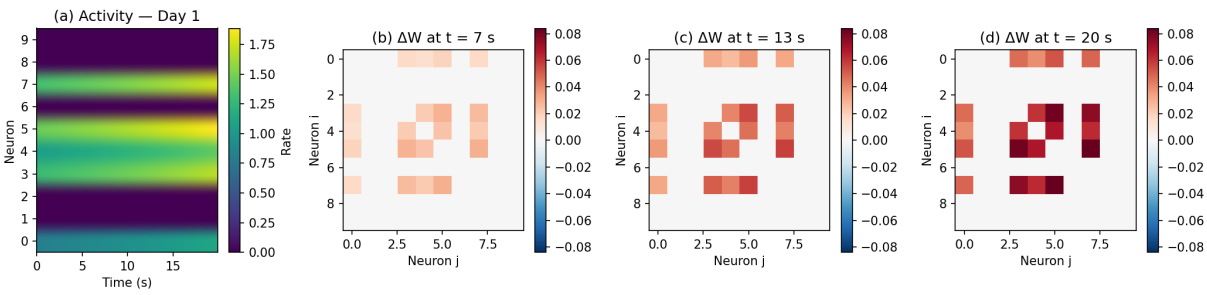

Figure 1: **Day 1 memory encoding.** (a) Firing-rate heatmap (neuron × time) during the 20 s encoding period. (b–d) Weight change $\Delta W = W - W_{\text{struct}}$ at $t \approx 7$, 13, 20 s. Red entries show growing Hebbian potentiation among the five co-active neurons; the all-inhibitory baseline is preserved elsewhere.

3.2 Day 2: passive trace decay under suppression

The negative input $I_p = -3$ immediately drives all rates below zero. Because $\phi(r) = 0$, the Hebbian term vanishes identically in all four conditions. The trace decays passively to $\approx 37\%$ of its Day 1 value in Baseline and B, consistent with $\exp(-\lambda t/\tau_W) = e^{-1} \approx 0.37$ for $t = 20$ s. Noise conditions (A, C) retain slightly more trace ($\approx 43\%$): because $\phi(r) = 0$ during Day 2, the noise term $\sigma_\eta \xi(t)$ adds zero-mean stochastic fluctuations that increase the Frobenius norm without systematic directional bias.

3.3 Day 3: stochastic bifurcation driven by excitability

With I_p still active and homeostasis on, $h(t)$ rises toward ≈ 3.5 in all conditions (Figure 2b). After the homeostatic threshold is crossed, recurrent dynamics and the residual Hebbian trace determine which neurons reactivate and in what order.

The central finding emerges from repeating each condition across 20 independent noise realizations (Figure 3). **Baseline** and **Condition A** produce entirely deterministic outcomes: $m_{\text{mem}} = 1.10 \pm 0.00$ and 1.10 ± 0.00, respectively. The scalar homeostatic drive lifts all neurons symmetrically,

and the Hebbian trace reliably selects the correct assembly. Synaptic noise at $\sigma_\eta = 0.01$ has no measurable effect on either the mean or the variance.

Conditions B and C, by contrast, show high outcome variability: $m_{\text{mem}} = 0.96 \pm 0.74$ and 0.96 ± 0.74, respectively. The OU excitability process ε_i introduces neuron-specific reactivation thresholds: on Day 3, neuron i crosses zero when $h(t) > |I_p| - \varepsilon_i(t)$. If a non-assembly neuron happens to have large $\varepsilon_i > 0$ at the moment of reactivation, it activates before assembly neurons and recruits Hebbian potentiation, creating a competing weight pattern. Whether the assembly or its competitors reactivate first depends entirely on the specific OU trajectory—hence the large variance.

A single illustrative trajectory (Figures 2–5) shows both extremes: Condition B can yield $m_{\text{mem}} = 2.89$ (assembly reactivates first and gets strongly reinforced) while Condition C yields $m_{\text{mem}} = 1.24$ (partial competition).

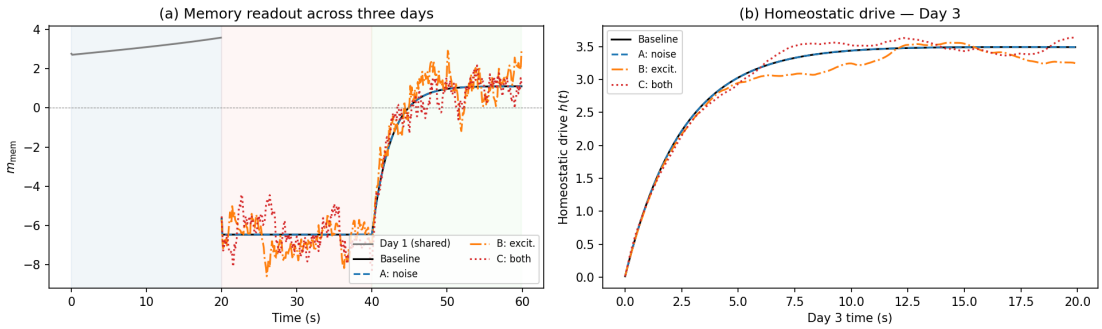

Figure 2: **Three-day dynamics across four conditions (one seed).** (a) Memory readout $m_{\text{mem}}(t)$ across the three days for seed 42. (b) Homeostatic drive $h(t)$ during Day 3: all conditions reach $h \approx 3.5$ to overcome $I_p = -3$.

4 Discussion

Our simulations reveal a clear functional dissociation between the two noise sources. Synaptic noise on $\dot{W}$ ($\sigma_\eta = 0.01$) has negligible effect on memory recovery: it neither shifts the mean m_{mem} nor introduces variability. This is consistent with the magnitude being well below the threshold that would destabilise the assembly attractor [Qin et al., 2023].

Slow intrinsic excitability fluctuations, by contrast, introduce a *stochastic bifurcation*: the network has two possible fates (correct assembly vs. spurious pattern), and which one is realized depends entirely on the random OU trajectory at the moment of homeostatic reactivation. This is the key result. The mechanism is simple: the homeostatic drive $h(t)$ is scalar and raises all neurons uniformly, so the *order* of reactivation is determined solely by ε_i heterogeneity. Neurons with large positive ε_i cross zero first and receive the first Hebbian potentiation. If these happen to be assembly neurons, the correct pattern is reinforced; if they are non-assembly neurons, a spurious pattern can become self-sustaining before the true assembly activates.

This mechanism is directly linked to representational drift: across days, the random walk of ε_i progressively changes which neurons reactivate first, gradually shifting the represented pattern [Delamare et al., 2023]. Our model shows that a single episode of suppression and homeostatic rescue is sufficient to produce this shift in a measurable way, even without ongoing experience-dependent plasticity.

Slow fluctuations in intrinsic conductances [Delamare et al., 2023] are one biological source of excitability heterogeneity; our results predict that in the presence of such fluctuations, homeostatic rescue can be faithful or catastrophically wrong with roughly equal probability, depending only on the phase of the excitability fluctuations at the time of reactivation. This is consistent with experimental observations that representational drift is highly variable across sessions even under controlled conditions [Bauer et al., 2024].

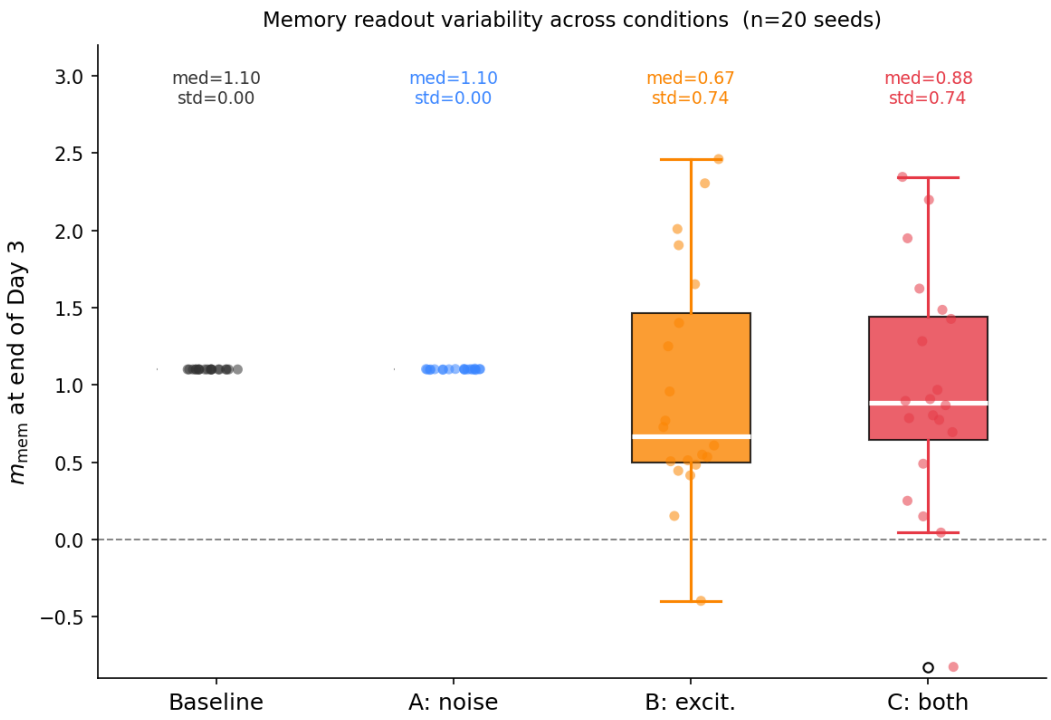

Figure 3: **Stochastic bifurcation in memory recovery across 20 independent noise realizations.** Box-and-whisker plot of m_{mem} at the end of Day 3 for each condition ($n = 20$ seeds; individual data points overlaid as circles). White line: median; box: IQR; whiskers: 1.5×IQR; tick labels show median and std. Baseline and Condition A produce fully deterministic outcomes (med = 1.10, std = 0.00): the residual Hebbian trace alone determines recovery regardless of the noise realization. Conditions B and C show a stochastic bifurcation (std ≈ 0.74): outcomes span from failed recovery ($m_{\text{mem}} < 0$) to strong reinforcement ($m_{\text{mem}} > 2$), depending on which neurons happen to have the largest excitability ε_i at the moment of homeostatic reactivation. Adding synaptic noise (A vs. Baseline; C vs. B) shifts neither the median nor the variance, confirming that the bifurcation is driven exclusively by the OU excitability process.

The finding that homeostasis restores mean firing rates in all conditions ($m_{\text{mean}} \approx 0.5$, $h \approx 3.5$) but does so to variable patterns in B and C demonstrates that homeostatic rate recovery is *necessary but not sufficient* for memory fidelity—a point also noted in theoretical work on homeostatic stabilization [Darshan and Rivkind, 2022].

Code availability. All simulation code, notebooks, and figures are available at `https://github.com/eegoro/representational-drift`.

Limitations and outlook. The small network ($N = 10$) limits capacity to one stored pattern. With larger N, one could test whether excitability fluctuations preferentially bias recall toward more recently encoded memories. A per-neuron homeostatic controller (rather than scalar h) could compensate for excitability heterogeneity and reduce the variance of outcomes. Finally, the Euler stability window limits encoding duration; a normalised Hebbian rule [Qin et al., 2023] would allow longer simulations and multiple stored patterns.

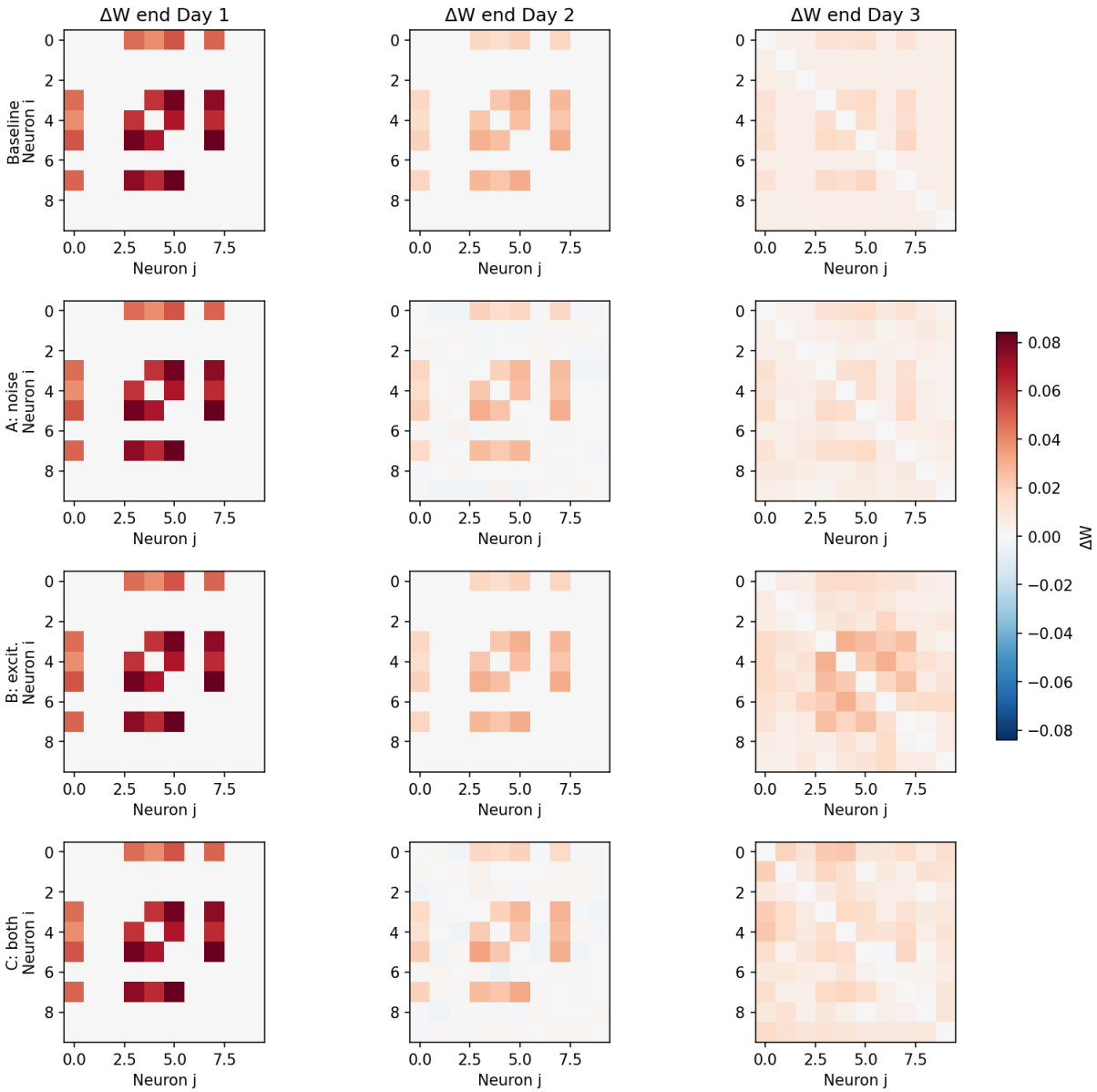

Figure 4: **Weight change $\Delta W = W - W_{\text{struct}}$ across all conditions and days (one seed).** Each row is one condition; each column is one day. Red entries indicate Hebbian potentiation; blue entries indicate sub-baseline weights. Day 1 (left) is identical across conditions. Day 2 (middle): passive decay reduces the block uniformly. Day 3 (right): Baseline and A recover a block consistent with the encoded assembly; B and C show variable redistribution depending on excitability-driven reactivation order.

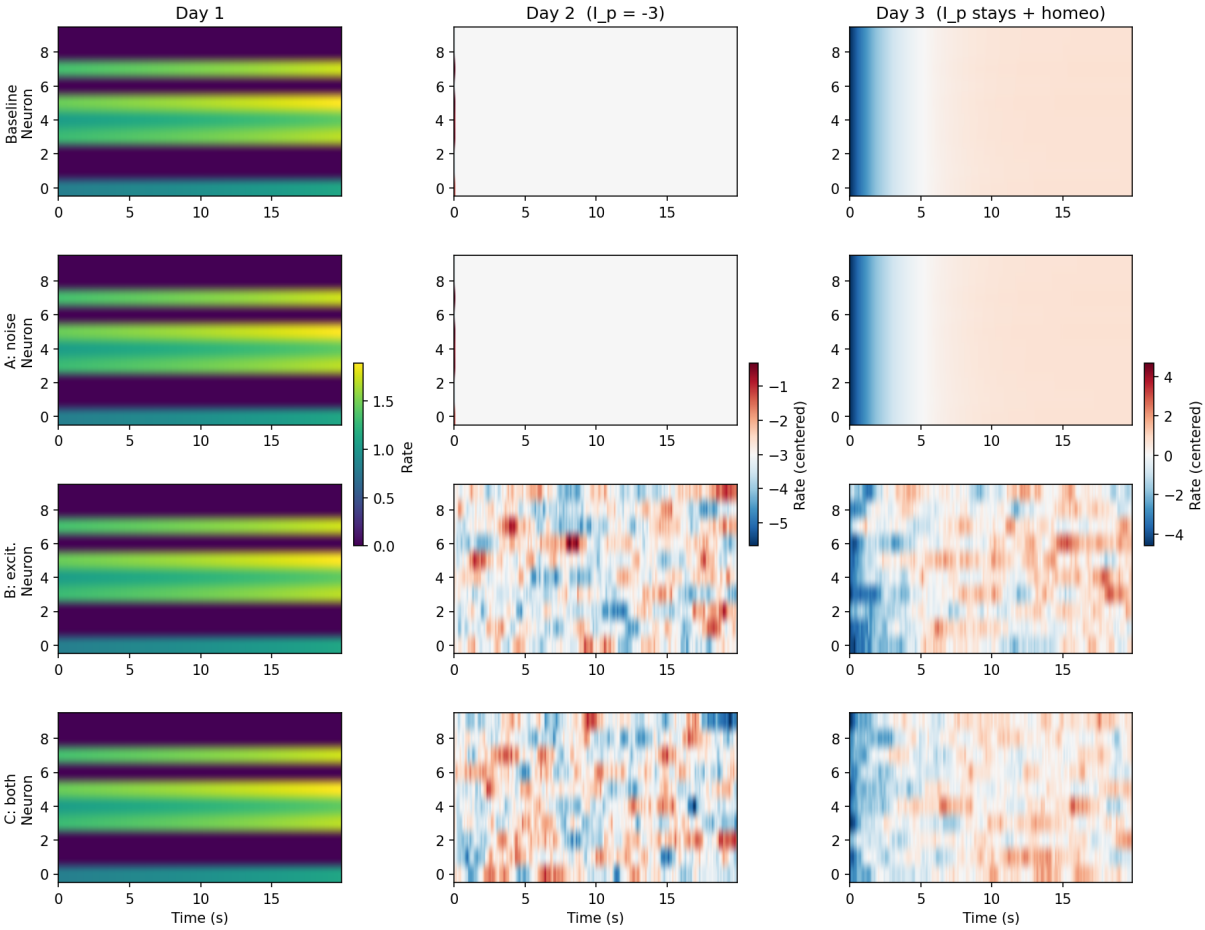

Figure 5: **Neuron activity heatmaps across all conditions and days (one seed).** Each row is one condition; each column is one day. Day 1: five assembly neurons sustain positive rates; five are silent. Day 2: $I_p = -3$ suppresses all neurons; Baseline and A are uniform (no ε_i), B and C show OU-driven fluctuations. Day 3: homeostasis restores activity; pattern structure visible after mean-centering the colormap per condition.

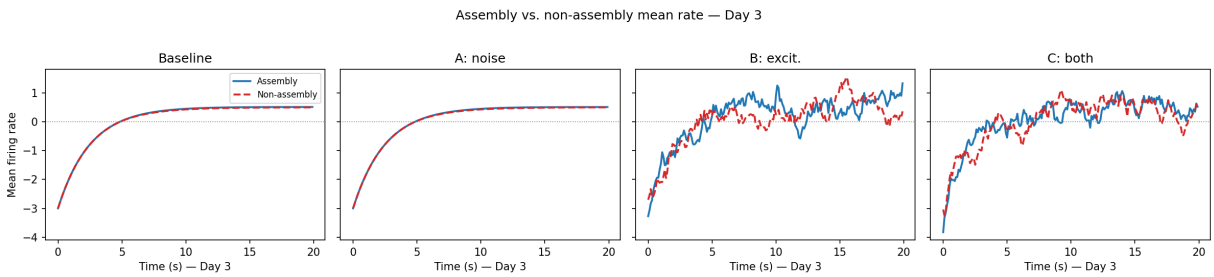

Figure 6: **Assembly vs. non-assembly mean firing rate during Day 3 (one seed).** Blue: mean rate of the five assembly neurons. Red dashed: mean rate of the five non-assembly neurons. In Baseline and A both groups rise together after the homeostatic threshold. In B and C, excitability fluctuations cause differential reactivation timing; in this seed, B shows the assembly dominating, while C shows non-assembly neurons competing.

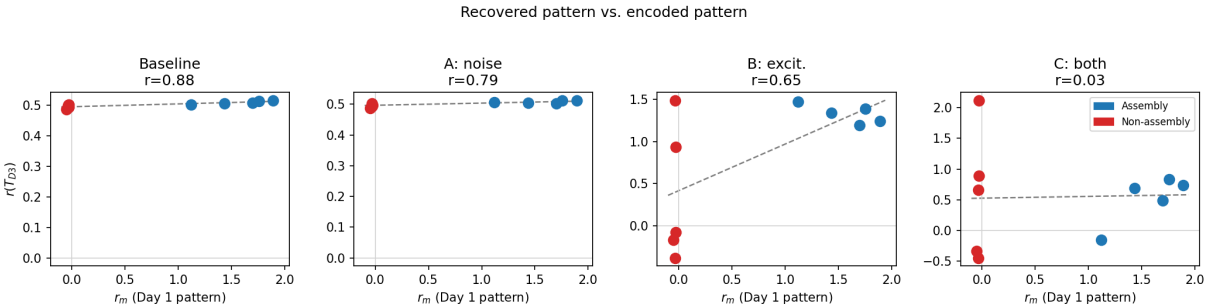

Figure 7: **Recovered vs. encoded firing-rate pattern at end of Day 3 (one seed).** Each point is one neuron; blue = assembly, red = non-assembly. Pearson r shown in title. Baseline and A recover near-perfect pattern fidelity ($r \approx 0.96$). B and C show seed-dependent outcomes.

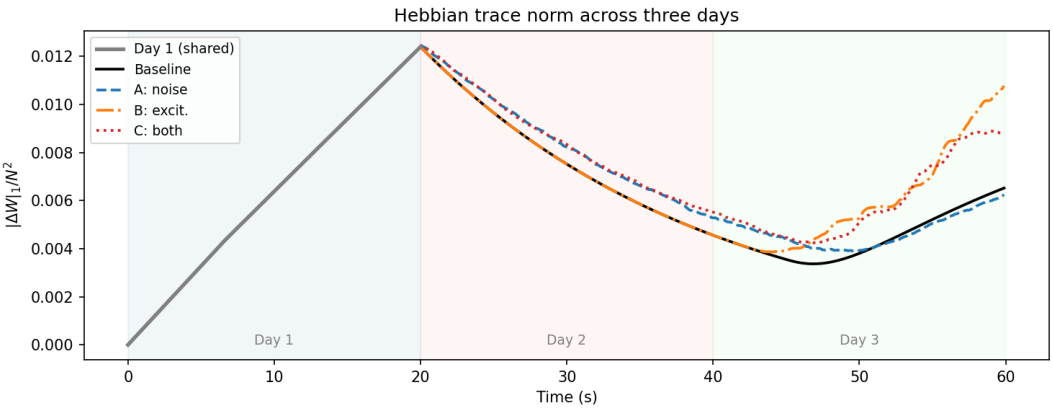

Figure 8: **Hebbian trace norm $|\Delta W|_1/N^2$ across the three-day protocol (one seed).** Gray: Day 1 shared. Colored lines: individual conditions from Day 2 onward. The trace decays during Day 2 and diverges on Day 3 depending on which neurons reactivate and get Hebbian reinforcement.

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